

**Problem 29 (JBMO Shortlist 2018 N2).** Solve the equation  $3^m + 271 = 125 \cdot 2^n$  in positive integers.

**Solution.** When  $n = 1, 2$  there is no solution. When  $n = 3, m = 6$  is a solution since  $729 + 271 = 125 \cdot 8$ . Suppose  $n \geq 4$ . Then we have  $3^m + 271 \equiv 0 \pmod{16}$ , which implies  $3^m \equiv 1 \pmod{16}$ . Since the order of 3 modulo 16 is 4, this yields  $4 \mid m$ . By the Fermat little theorem, we have  $3^m + 271 \equiv 1 + 1 = 2 \pmod{5}$ , contradicting  $5 \mid 125 \cdot 2^n$ . So  $(m, n) = (6, 3)$  is the only solution.  $\square$

**Problem 30 (JBMO 2019 Problem 4).** There is a  $5 \times 100$  table. We choose  $n$  of the cells and call them *special cells*. If every cell has at most 2 adjacent special cells (cells sharing common sides), find the greatest possible value of  $n$ .

**Solution.** The answer is 302.

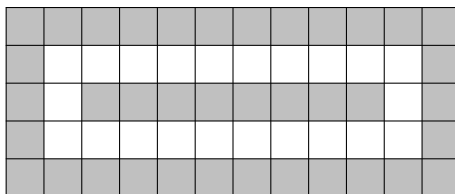
We first prove that  $n \leq 302$ . We count the number  $N$  of pairs of adjacent cells  $(A, B)$  such that  $A$  is special (and  $B$  can be special or not). For each cell  $B$ , there are at most two choices of  $A$ . Therefore, we have  $N \leq 500 \times 2 = 1000$ . For each special cell  $A$ , the number of choices of  $B$  is the number of adjacent cells of  $A$ . If there are  $x$  special cells at the corners,  $y$  special cells on the edges (but not at the corners), and  $z$  special interior cells, then we have  $N = 2x + 3y + 4z$ . It follows that  $2x + 3y + 4z \leq 1000$ .

Since there are only 4 corner cells, and  $500 - 3 \times 98 - 4 = 202$  cells on the edges, we must have  $x \leq 4$  and  $y \leq 202$ . Therefore, we see that

$$4(x + y + z) = (2x + 3y + 4z) + 2x + y \leq 1000 + 2(4) + (202) = 1210.$$

As  $n = x + y + z$  is an integer, we must have  $n \leq 302$ .

Lastly, we can choose all corner cells and all cells on the edges, together with the 96 cells not adjacent to any of the boundary cells. (The figure below shows an analogous case for a  $5 \times 12$  board.) There are  $500 - 3 \times 98 + 96 = 302$  chosen cells in total, and it is not hard to see that the condition is satisfied. So the greatest possible value of  $n$  is 302.



$\square$

**Problem 31 (Dutch TST 2016 Day 1 Problem 4).** Find all functions  $f : \mathbb{R} \rightarrow \mathbb{R}$  satisfying

$$f(xy - 1) + f(x)f(y) = 2xy - 1 \tag{1}$$

for all  $x, y \in \mathbb{R}$ .

**Solution.** The solutions are  $f(x) = x$  for any  $x \in \mathbb{R}$ , and  $f(x) = -x^2$  for any  $x \in \mathbb{R}$ . The former is obviously a solution. Here we check the latter as follows.

$$f(xy - 1) + f(x)f(y) = -(xy - 1)^2 + (-x^2)(-y^2) = -x^2y^2 + 2xy - 1 + x^2y^2 = 2xy - 1.$$

Putting  $x = 0$  in (1), we obtain  $f(-1) + f(0)f(y) = -1$ . If  $f(0) \neq 0$ , this implies  $f(y)$  is a constant. But then (1) cannot hold for all  $x, y \in \mathbb{R}$ . Therefore, we must have  $f(0) = 0$  and  $f(-1) = -1$ . Putting  $x = y = 1$  in (1), we obtain  $f(1)^2 = 1$ .

**Case 1:**  $f(1) = 1$

Putting  $y = 1$  in (1), we get

$$f(x-1) + f(x) = 2x - 1. \quad (2)$$

Replacing  $x$  by  $xy$ , this becomes  $f(xy-1) + f(xy) = 2xy - 1$ . Comparing this with (1), we see that

$$f(xy) = f(x)f(y). \quad (3)$$

Now, by (3), we know that  $f(x^2-1) = f(x-1)f(x+1)$ . We can use (2) to replace each term to  $f(x^2) = f(x)^2$  and  $f(x)$ . Indeed, we have

$$\begin{aligned} & f(x^2-1) = f(x-1)f(x+1) \\ \Rightarrow & 2x^2 - 1 - f(x^2) = [2x - 1 - f(x)][2x + 1 - f(x)] \\ \Rightarrow & 2x^2 - 1 - f(x)^2 = f(x)^2 - 4xf(x) + 4x^2 - 1 \\ \Rightarrow & 2f(x)^2 - 4xf(x) + 2x^2 = 0 \\ \Rightarrow & f(x) = x. \end{aligned}$$

**Case 2:**  $f(1) = -1$

Putting  $y = 1$  in (1), we get

$$f(x-1) - f(x) = 2x - 1. \quad (4)$$

Replacing  $x$  by  $xy$ , this becomes  $f(xy-1) - f(xy) = 2xy - 1$ . Comparing this with (1), we see that

$$f(xy) = -f(x)f(y). \quad (5)$$

Now, by (5), we know that  $f(x^2-1) = -f(x-1)f(x+1)$ . Using (4), we have

$$\begin{aligned} & f(x^2-1) = -f(x-1)f(x+1) \\ \Rightarrow & f(x^2) + 2x^2 - 1 = -[f(x) + 2x - 1][f(x) - 2x - 1] \\ \Rightarrow & -f(x)^2 + 2x^2 - 1 = -f(x)^2 + 2f(x) + 4x^2 - 1 \\ \Rightarrow & 2f(x) + 2x^2 = 0 \\ \Rightarrow & f(x) = -x^2. \end{aligned}$$

This completes the proof. □

**Problem 32 (China TST 2015 Test 2 Problem 3).** Let  $\triangle ABC$  be an acute triangle with circumcentre  $O$  and centroid  $G$ . Define the following.

- $D$  is the midpoint of  $BC$
- $E$  is a point inside  $\triangle ABC$  such that  $AE \perp BC$  and  $\angle BEC = 90^\circ$
- $F$  is the intersection of  $EG$  and  $OD$
- $P$  is a point on  $BC$  such that  $FP \parallel OB$
- $Q$  is a point on  $BC$  such that  $FQ \parallel OC$
- $X$  is a point on  $AB$  such that  $XP \perp BC$
- $Y$  is a point on  $AC$  such that  $YQ \perp BC$
- $\Gamma$  is the circle tangent to  $OB$  and  $OC$  at  $B$  and  $C$  respectively
- $\Omega$  is the circumcircle of  $\triangle AXY$

Prove that  $\Gamma$  and  $\Omega$  are tangent to each other.

